THE FINITE PART ONLY SEES THE LOCALLY QUASI-FINITE LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers within this span.

Allowed standard prerequisites (editorial).

- Canonical injection on finite parts. lemma-finite-open (same chapter, NOT carded, declared external). Gives (X'/Y)_fin -> (X/Y)_fin injective.
- The locally quasi-finite locus commutes with base change. Morphisms of Spaces, lemma locally-finite-type-quasi-finite-part (cross-chapter, named). Lets the argument work after arbitrary base change.
- Finite implies locally quasi-finite. Morphisms of Spaces, lemma finite-quasi-finite (cross-chapter, named). Closes the inclusion Z subset X'.

Source reference. Original lemma 04PF: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1628–1628. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NY: focused-seeds/geometry/S9-smg-finite-lives-on-locally-quasi-finite-part/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NW: focused-seeds/geometry/S9-smg-finite-lives-on-locally-quasi-finite-part/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $X' \subset X$ be the maximal open subspace over which f is locally quasi-finite, see Morphisms of Spaces, Lemma 04NW. Then $(X/Y)_{fin} = (X'/Y)_{fin}$.*

Proof. Lemma 04PF gives us an injective map $(X'/Y)_{fin} \to (X/Y)_{fin}$. Morphisms of Spaces, Lemma 04NW assures us that formation of X' commutes with base change. Hence everything comes down to proving that if $Z \subset X$ is an open subspace such that $f|_Z : Z \to Y$ is finite, then $Z \subset X'$. This is true because a finite morphism is locally quasi-finite, see Morphisms of Spaces, Lemma 04NY. □

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.